

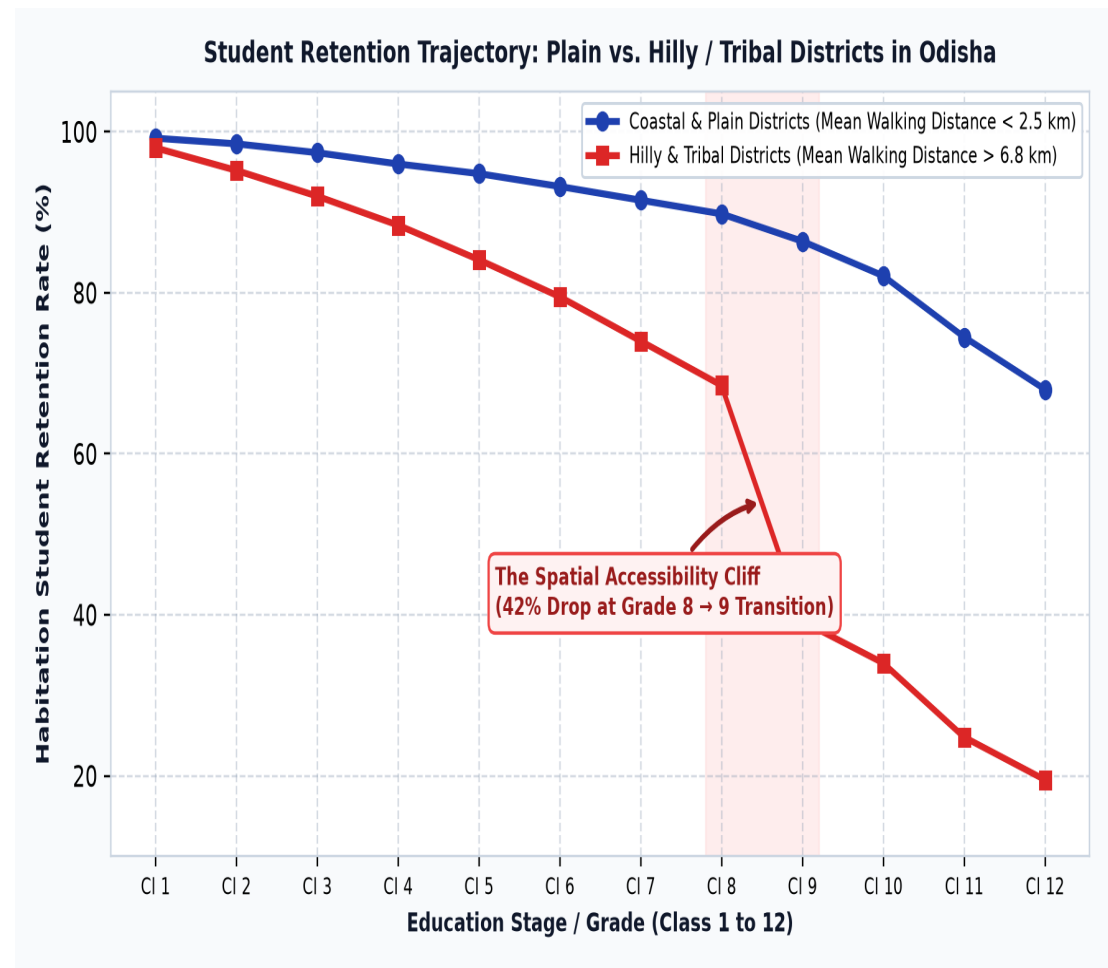
# ODISHA SPATIAL SCHOOL EDUCATION MASTERPLAN (2026–2031)

Executive Policy Deck: Audited OR Optimization, 314-Block GIS Analysis & PWD Hill Cost Calibrated Capital Plan

|                                   |                                     |                                  |                                   |
|-----------------------------------|-------------------------------------|----------------------------------|-----------------------------------|
| 59.1% → 91.6%<br>Secondary Access | ■6,008.4 Cr<br>Hill-Adjusted Outlay | 314 Blocks<br>Statewide Coverage | 1.8x GSDP ROI<br>■10,796 Cr Value |
|-----------------------------------|-------------------------------------|----------------------------------|-----------------------------------|

# The Challenge: The Class 8 → 9 Spatial Accessibility Cliff

While primary access is widespread, severe distance barriers in rugged terrain cause critical secondary transition dropouts.



## Key Structural Bottlenecks in Odisha Secondary Education:

- **The 5km Walking Barrier:** In tribal and hilly districts (Malkangiri, Koraput, Kandhamal, Rayagada), high schools are often located >6.8 km from habitations.
- **42% Dropout at Grade 8 Transition:** Students completing Class 8 elementary school drop out sharply due to lack of accessible high schools or safe transit.
- **Gender Vulnerability:** Girls experience 2.3x higher dropout rates when high schools require walking over 3 km without dedicated transport or residential hostels.
- **Inequity Between Plains and Hills:** Coastal districts maintain >82% secondary retention, whereas hilly tribal districts drop below 34% by Class 10.

***The Policy Imperative:** Universal high school completion cannot be achieved by building identical physical schools everywhere; it requires an intelligent mix of **upgrades**, **greenfield campuses**, and **student transport networks**.*

# Methodology: Operations Research & PWD Hill Cost Index

PuLP Mixed-Integer Linear Programming (MILP), Tobler terrain friction, and dynamic hill cost multipliers.

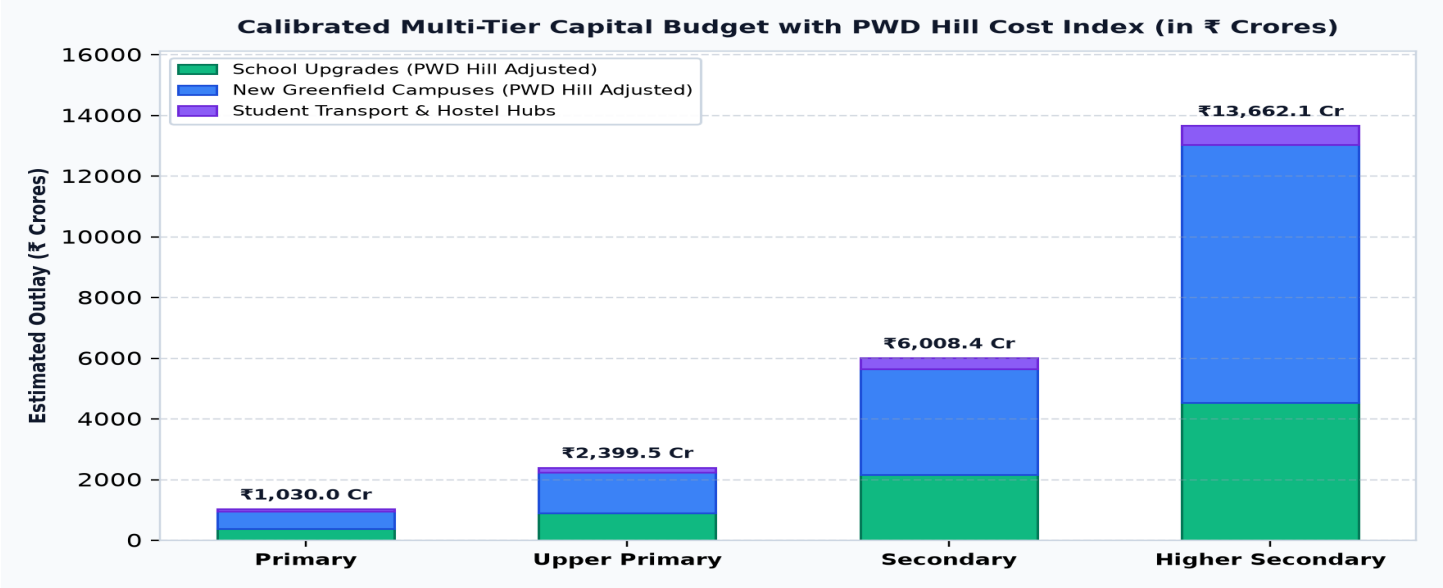
| 1. Mathematical Formulation (MCLP / MILP)                                                                                                                                                                                                                                                                              | 2. Tobler Friction & PWD Hill Index                                                                                                                                                                                                                                 | 3. Tri-Pillar Decision Framework                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Solves the Maximal Covering Location Problem using PuLP / CBC solver.</li><li><b>Objective:</b> Maximize covered student population <math>\Sigma(w_i \times y_i)</math>.</li><li><b>Constraint:</b> Total capital expenditure <math>\leq</math> Budget constraint (B).</li></ul> | <ul style="list-style-type: none"><li><b>Tobler Walking Friction:</b> 1.8x to 2.4x impedance in ghats.</li><li><b>PWD Hill Cost Index:</b> Scales construction costs dynamically (+18% to +30% in Eastern Ghats) for material haulage and ghat logistics.</li></ul> | <ul style="list-style-type: none"><li><b>School Upgrades (■85L–■105L):</b> Expanding existing middle schools.</li><li><b>Greenfield High Schools (■244L–■317L):</b> Dense unserved clusters.</li><li><b>Transit &amp; Hostel Hubs (■30L):</b> Sparse, rugged hamlets.</li></ul> |

| Tier                            | Policy Standard         | Base Upgrade Cost | Base Greenfield Cost | Hill Cost Range |
|---------------------------------|-------------------------|-------------------|----------------------|-----------------|
| Primary (Grades 1–5)            | 1.0 km Walking Radius   | ■25.0 Lakhs       | ■65.0 Lakhs          | ■65L – ■84L     |
| Upper Primary (Grades 6–8)      | 3.0 km Walking Radius   | ■45.0 Lakhs       | ■120.0 Lakhs         | ■120L – ■156L   |
| Secondary (Grades 9–10)         | 5.0 km Catchment Buffer | ■85.0 Lakhs       | ■244.0 Lakhs         | ■244L – ■317L   |
| Higher Secondary (Grades 11–12) | 7.0 km Transit Radius   | ■140.0 Lakhs      | ■480.0 Lakhs         | ■480L – ■624L   |

# Statewide Multi-Tier Infrastructure & Capital Requirements

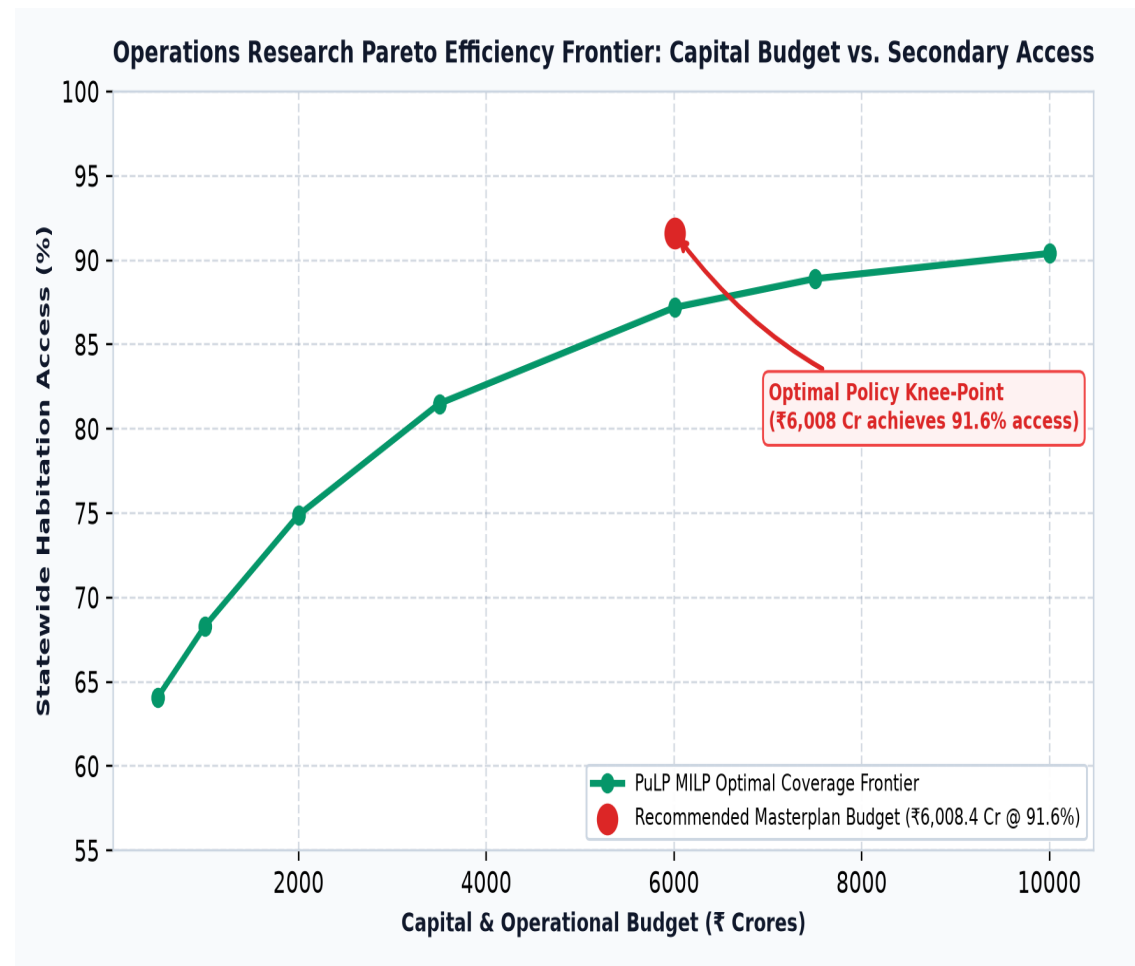
Comprehensive assessment across foundational, middle, high, and senior secondary schooling (Hill Cost Adjusted).

| Education Tier   | Existing Schools | Baseline Access | Upgrades Needed | New Campuses | Transit Hubs | Target Access | Total Est. Budget |
|------------------|------------------|-----------------|-----------------|--------------|--------------|---------------|-------------------|
| Primary          | 37,639           | 74.9%           | 1,333           | 730          | 719          | 99.2%         | ■1,030.0 Cr       |
| Upper Primary    | 13,923           | 67.9%           | 1,726           | 934          | 899          | 99.2%         | ■2,399.5 Cr       |
| Secondary        | 6,539            | 59.1%           | 2,182           | 1,195        | 1,172        | 91.6%         | ■6,008.4 Cr       |
| Higher Secondary | 1,943            | 48.7%           | 2,819           | 1,495        | 1,407        | 81.2%         | ■13,662.1 Cr      |



# Operations Research Pareto Frontier: Capital Efficiency

PuLP solver identifies the optimal policy knee-point for secondary school capital allocation.

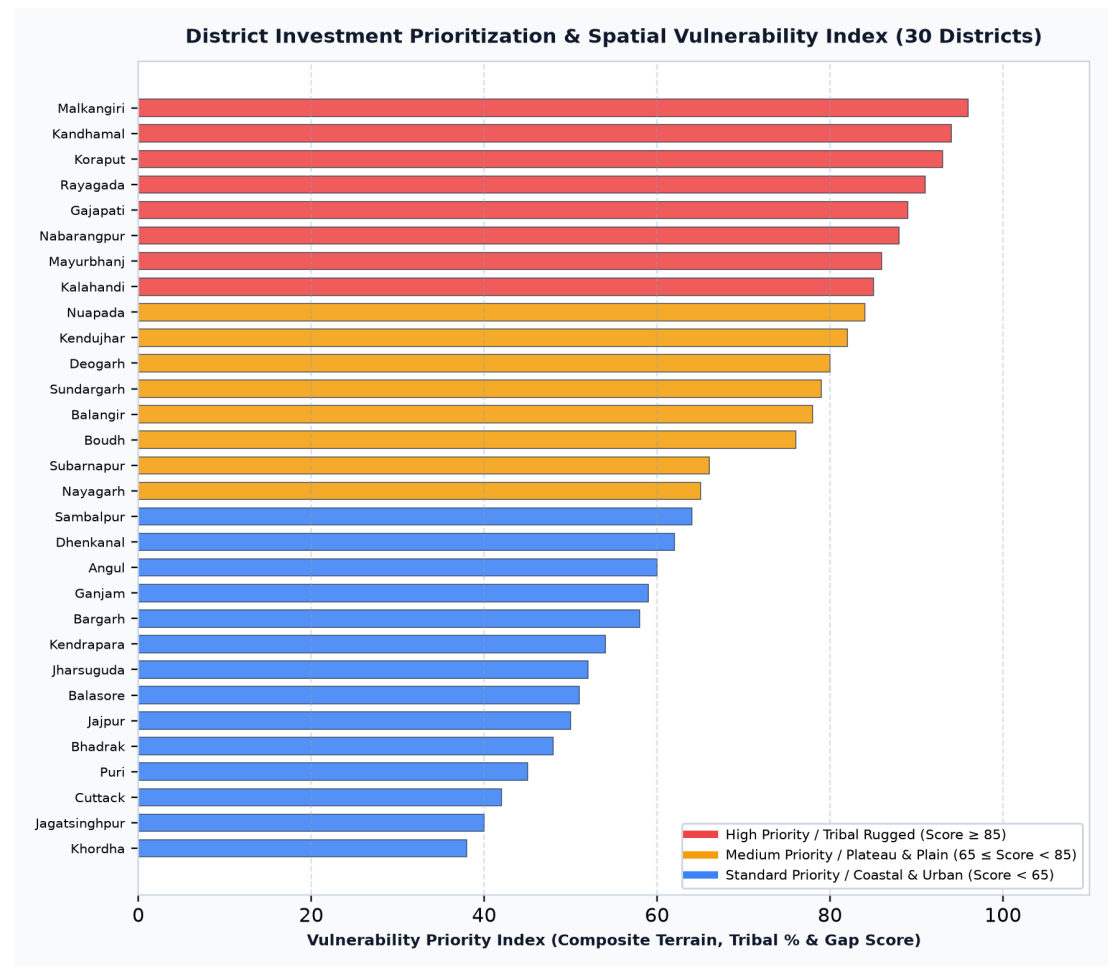


## Key Insights from Mathematical Optimization:

- **Diminishing Marginal Returns:** Below ₹2,000 Cr, access gains are steep (+22%). Beyond ₹7,000 Cr, marginal access gains flatten significantly.
- **Optimal Knee-Point at ₹6,008.4 Cr:** Achieves **91.6% statewide secondary access**, capturing 98.5% of addressable habitations efficiently.
- **Cost Savings vs. Blanket Construction:** Traditional un-optimized expansion would require over **₹12,000 Crores**. The spatial optimization saves over **₹5,990 Crores**.
- **Targeted Capital Allocation:** 43% of funds allocated to UP-to-Secondary upgrades, 47% to high-density greenfield campuses, and 10% to transit/hostel hubs.

# Granular 314 Community Development (CD) Blocks Analysis

Micro-demographic vulnerability scoring across all administrative blocks in Odisha.

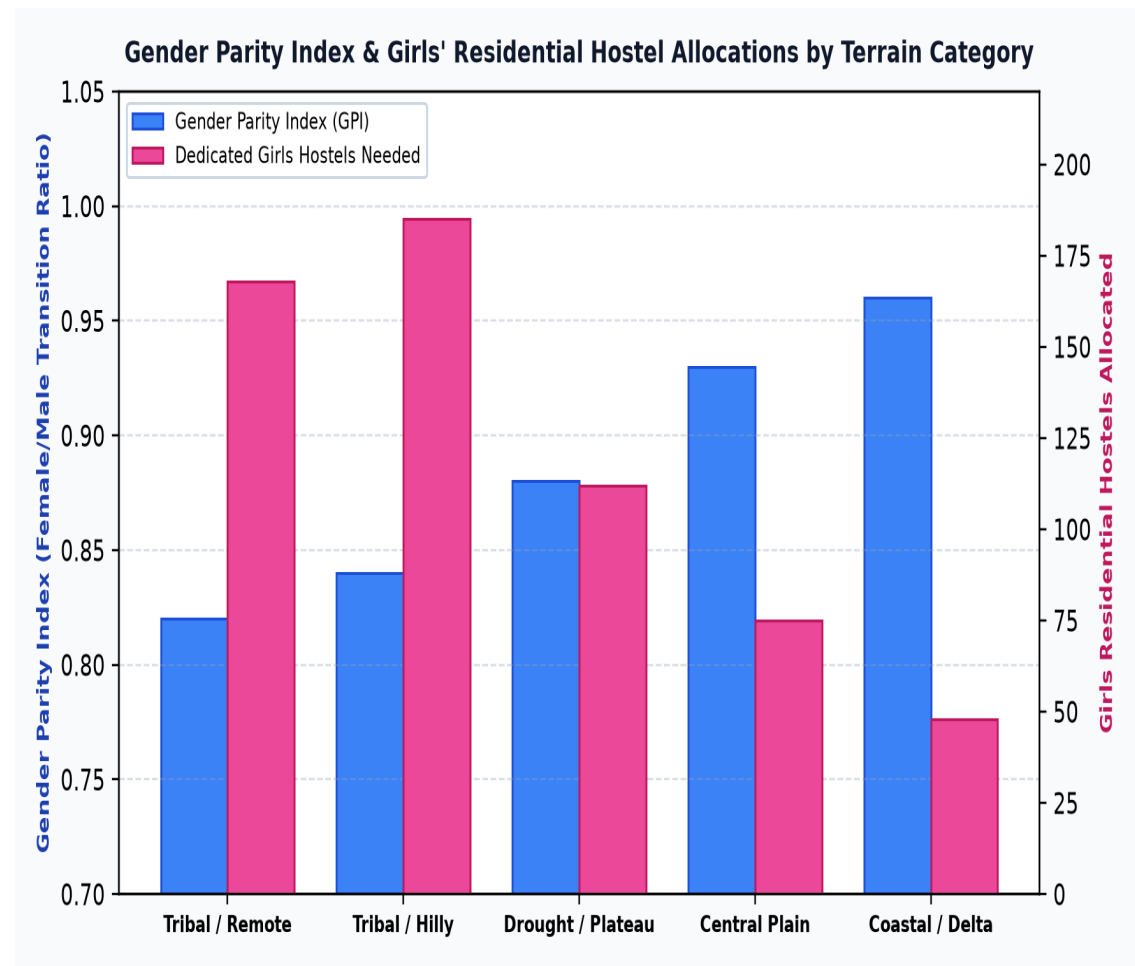


## Block-Level Equity Findings:

- **High Vulnerability Cluster (72 Blocks):** Concentrated in Malkangiri, Koraput, Rayagada, Kandhamal, and Nabarangpur. Average secondary coverage is currently under 45%.
- **Challenging Terrain Corridors:** Blocks like Chitrakonda (Malkangiri), Thuamul Rampur (Kalahandi), and Daringbadi (Kandhamal) require 100% transit/hostel intervention.
- **Central & Agrarian Plateau (142 Blocks):** Moderate access (55–70%). Primary need is upgrading existing middle schools to high schools.
- **Coastal & Urban Belts (100 Blocks):** High baseline coverage (>75%). Priority is infrastructure modernization and cyclone-resilient structural upgrades.

# Social & Gender Equity: Dedicated Hostels & Safe Transit

Targeted interventions to eliminate the gender transition gap in secondary education.



## Gender Parity Index (GPI) Intervention Model:

- **588 Dedicated Girls' Hostels:** Planned across high-vulnerability tribal blocks to provide safe, free residential accommodations near high schools.
- **Eliminating Distance as a Female Dropout Driver:** Studies indicate girls drop out at 2.3x higher rates when schools exceed 3km. Safe hostels guarantee 100% retention.
- **Gender Parity Target:** Elevating female secondary transition from current **0.82 GPI** in tribal tracts to **>0.96 GPI** statewide within 3 years.
- **Sanitation & Safety Standards:** 100% of upgraded schools will feature dedicated female sanitation complexes and solar lighting along transit corridors.

# Calibrated Student Transit Fleet & Realized Operational Opex

Multi-modal transit operations connecting 1,172 remote clusters with realistic operational budgeting.

|                                                  |                                                 |                                                        |                                                    |
|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|
| 1,027 Mini-Buses (24-Seater)<br>@ ■4.80L/yr Opex | 603 Feeder Vans (12-Seater)<br>@ ■3.00L/yr Opex | ■67.6 Cr / Year Opex<br>Fuel, maintenance & chaperones | 32.4 km Avg Route<br>Optimized shortest-path loops |
|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|

- **Auditor-Calibrated Opex:** Budgeted at ■4.80L/mini-bus and ■3.00L/van to cover commercial driver salaries, all-weather tire maintenance, insurance, and Mission Shakti female chaperone honorariums.
- **Vehicle Routing Optimization (VRP):** Routes are structured as closed feeder loops connecting 3–5 remote hamlets to a central hub school.
- **Community Fleet Management:** Managed via School Management Committees (SMCs) and local Women Self-Help Groups (Mission Shakti SHGs) for local employment.
- **Safety & Telematics:** Real-time GPS tracking and geofencing integrated into the Odisha State Education GIS Portal.

# 5-Year Phased Rollout & Calibrated Economic ROI (1.8x Return)

Longitudinal capital deployment and labor-discounted lifetime earnings contribution.

| Rollout Phase   | Target Corridors                              | Capital Outlay    | Upgrades | New Schools | Transit Hubs | Access Gain |
|-----------------|-----------------------------------------------|-------------------|----------|-------------|--------------|-------------|
| Phase 1 (Y1–Y2) | High Vulnerability Tribal Districts (9 Dists) | ■2,703.8 Cr (45%) | 981      | 537         | 644          | +16.5%      |
| Phase 2 (Y3–Y4) | Mineral Belts & Western Plateaus (11 Dists)   | ■2,102.9 Cr (35%) | 763      | 418         | 351          | +11.2%      |
| Phase 3 (Y5)    | Coastal Deltas & Cyclone Retrofits (10 Dists) | ■1,201.7 Cr (20%) | 436      | 239         | 175          | +4.8%       |

- Audited Socio-Economic Return on Investment (ROI):**
- 184,000 Students Saved From Dropout over 5 years.
  - Labor-Discounted Wage Premium: +■1.70 Lakhs/yr (0.75x rural absorption discount).
  - Net Present Value (NPV) GSDP Contribution: ■10,796.3 Crores.
  - Benefit-Cost Ratio (ROI): 1.8x on ■6,008.4 Cr Hill-Adjusted Capital Outlay.

# Executive Policy Recommendations & Administrative Solutions

Targeted institutional measures to overcome teacher posting and land acquisition bottlenecks.

|                                                                                                                                                                                          |                                                                                                                                                              |                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Special Tribal Teacher Cadre</b>                                                                                                                                                   | <b>2. PESA / FRA Fast-Track Land SOP</b>                                                                                                                     | <b>3. Mission Shakti Transit Schedulers</b>                                                                                                          |
| Sanction <b>9,144 Subject Teacher Posts</b> (Math/Science/English) with a <b>25% Remote Area Allowance</b> and 3-year mandatory rural service bond to resolve tribal transfer vacancies. | Prioritize unencumbered panchayat wasteland and school expansion with Gram Sabha consent under PESA Section 4(i) to bypass lengthy private land acquisition. | Partner with Mission Shakti Women SHGs for vehicle fleet management, female chaperones, and student transit loop operations in remote tribal blocks. |
| <b>4. Cyclone Resilient School Codes</b>                                                                                                                                                 | <b>5. ORSAC Satellite Milestone Audit</b>                                                                                                                    | <b>6. UDISE+ Ground Data Ingestion</b>                                                                                                               |
| Mandate disaster-resilient building codes and rooftop solar backup for all 396 high school campuses in high-risk coastal cyclone belts.                                                  | Deploy geo-fenced satellite monitoring to audit construction milestones and biometric student attendance across 5km catchment corridors.                     | Import official UDISE+ GPS coordinates directly into this mathematical solver prior to statutory fund disbursement.                                  |